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PERSONAL INFORMATION

Pengpei Zhu (朱芃佩)

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Nationality Chinese

EDUCATION

Nov. 2024 – Present

Ph.D. in Astrophysics

Cosmic Dawn Center (DAWN), Denmark

Technical University of Denmark (DTU), Kgs. Lyngby, Denmark

Sep. 2022 – Sep. 2024

MSc. in Astronomy (Astronomy Research)

Leiden University, Leiden, The Netherlands

– Grade: 8.2/10
– 2024 KHMW Young Talent Awards in Astronomy (3000 EUR)

Sep. 2017 – Mar. 2022

B.S. in Physics (Astrophysics) and B.A. in Philosophy

University of California at Santa Cruz (UCSC), Santa Cruz, California, USA

– GPA: 3.51/4.0
– With the Highest Dean's Scholarship (40000 USD)

RESEARCH EXPERTISE & INTERESTS

– High-redshift galaxies, galaxy quenching, galactic chemical enrichment, galaxy dynamics, galaxy interactions, galaxy formation & assembly;

– Photometry, near-infrared spectroscopy, integral-field spectroscopy;

– AI for science, data availability, package/software development

MAIN PUBLICATIONS

[1] **Pengpei Zhu**, Kei Ito, Francesco Valentino, et al. "There and Back Again: Neutral Outflows in z $\sim$ 3.5 Quiescent Galaxies". In: *Astronomy & Astrophysics* 712 (Aug. 2026), A169. DOI: 10.1051/0004-6361/202659510.

[2] **Pengpei Zhu**, Katherine A. Suess, Mariska Kriek, et al. "SQuIGG$\vec{L}$E: Observational Evidence of Low Ongoing Star Formation Rates in Gas-rich Post-starburst Galaxies". In: *The Astrophysical Journal* 981 (Mar. 2025), p. 60. DOI: 10.3847/1538-4357/adaf1c.

[3] Min Bao, Yanmei Chen, **Pengpei Zhu**, et al. "Different Formation Scenarios for Counterrotating Stellar Disks in Nearby Galaxies". In: *The Astrophysical Journal Letters* 926.2 (Feb. 2022), p. L13. DOI: 10.3847/2041-8213/ac52ad.

CONFERENCES AND WORKSHOPS

Jun. 29 – Jul. 3, 2026 **EAS Annual Meeting 2026**: ePoster
There and back again? Neutral outflows in $z \sim 3.5$ quiescent galaxies

Jun. 8 – 12, 2026 **GalFirenze 2026**: Flash talk (5 min)

Apr. 13 – 17, 2026 **Charting Cosmic Dawn in Copenhagen**: Contributed talk (15 min)
DeepDive: Neutral gas outflows in $z \sim 3.5$ quiescent galaxies

Mar. 23 – 27, 2026 **Tokyo JWST 2026**: Contributed talk (7 min)
DeepDive: Neutral gas outflows in $z \sim 3.5$ quiescent galaxies

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Aug. 16 – 22, 2025 **NOT Summer School 2025**: Summer school about design/practice observations with the Nordic Optical Telescope at La Palma, Spain for 1 week.

May 20 – 21, 2025 **Annual Danish Astronomical Meeting 2025**: Poster
Neutral gas outflows in $z \sim 3.5$ quiescent galaxies

AWARDS AND HONORS

KHMW Young Talent Awards in Astronomy: Royal Holland Society of Sciences and Humanities (KHMW) Master Thesis Prize **(3000 EUR)**, presented 2 in astronomy annually in the Netherlands. Click here for the Jury Report.

UCSC Dean's Scholarship: The highest for undergraduate international students (**40000 USD**)

UCSC Dean's Honor List: Winter 2020, Fall 2018, Winter 2018, Winter 2020, Spring 2021

RECENT RESEARCH PROJECTS

Oct. 2025 – Present (Ph.D. & Research Visit)

Stellar Metallicity of High-z Massive Quiescent Galaxies

INAF – Arcetri Astrophysical Observatory, Florence, Italy

Cosmic DAWN Center, DTU Space, Lyngby, Denmark

Collaborators: Dr. Anna Gallazzi, Dr. Stefano Zibetti, Dr. Francesco Valentino, Dr. Kei Ito

- We are studying a large sample of **100** $1 < z < 4.6$ **massive quiescent galaxies with JWST/NIRSpec data**, to understand their **stellar metallicity** with absorption lines.
- Our method is consistent with two published large local SDSS and LEGA-C studies, enabling us to compare the high-z sample with local samples without model discrepancies.
- **Two papers in a series are in prep. for *A&A*.**

Nov. 2024 - Feb. 2026 (Ph.D.)

Neutral Gas Outflow in z>3 Massive Quiescent Galaxies with JWST

Cosmic DAWN Center, DTU Space, Lyngby, Denmark

Supervisor: Dr. Francesco M. Valentino

- I worked with the JWST/NIRSpec+NIRCam program `DeepDive`, to study massive ($M_\star \sim 10^{11}\,M_\odot$) quiescent galaxies at z>3 for their neutral gas outflows with NaI D.
- **Publiushed on *A&A*** [1].

Aug. 2023 – Aug. 2024 (MSc. Thesis)

NGC 833 & NGC 835 as a case study of galaxy interaction in compact groups seen by VLT/MUSE

Leiden Observatory, Leiden, the Netherlands

Advisor: Dr. Ana Monreal Ibero

- I worked with VLT/MUSE IFU data of NGC 833/835, two nearby interacting galaxies. I developed algorithms to **fit and semi-automatically determine multi-component emission-line models using spatial, physical, and statistical information**.
- **Master Thesis Awarded 2024 KHMW Young Talent Awards in Astronomy (3000 EUR).**

Sep. 2022 – Jul. 2023 (MSc.)

Post-Starburst Galaxies with Large Molecular Gas Reservoirs

Leiden Observatory, Leiden, the Netherlands

Advisor: Prof. Dr. Mariska Kriek

- We studied 13 post-starburst galaxies at $z \sim 0.7$ with considerable molecular gas reserves to measure their.
- **I reduced Keck/NIRES spectral data** with `PyPeIt` and self-developed scripts.
- **Published on *ApJL*** [2].

COLLABORATIONS & SERVICES

`DeepDive` Collaboration: (Nov. 2024 — Present) JWST GO PID #3567 (PI: Francesco Valentino) program on $z > 3$ quiescent galaxies, related publications [1, 7, 8, 10, 11, 13, 17].

MUSE Science Team: (Dec. 2024 — Jun. 2026) VLT/MSUE collaboration member.

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SQuIGGLE Collaboration: (Sep. 2022 — Present) Massive post-starburst galaxies at $z \sim 7$ with multiple ground/space-based observations (SDSS, DESI, Keck, Magellan, Gemini, HST, Subaru/HSC, ALMA, VLA, WISE, JWST/NIRSpec IFU), related publications [2, 5, 6, 14, 15].

Committee services: (Jan. 2026 — Dec. 2027) DTU Space PhD Committee member, in charge of practical matters related to the PhD School.

(Jun. 2025 — Jun. 2026) Cosmic Dawn Center Committee member, discussing DAWN-related practical matters and future strategy.

DAWN Summit 2025 Scientific Organizing Committee (SOC) member.

Outreach: (Oct. 2022 - Oct. 2023) Member of the observing committee in *L.A.D 'F. Kaiser'* Astronomy Club at Leiden, organizing public observing events and operating telescopes at the Leiden Observatory.

TECHNICAL STRENGTHS

Programming Skills: Python, IDL

Software & Packages: `pPXF, PypeIt, Bagpipes, DS9, LaTeX, FADO`

Observation Experiences: 5 nights at the Nordic Optical Telescope; 1 night at the Lick Observatory's 1-m Nickel Telescope and 3-m Shane Telescope.

Proposal Writing: PI on two submitted JWST proposals (not accepted), with JWST/NIRSpec MSA (4th Quantile in cycle 5) and with JWST/NIRSpec IFU (cycle 6).

Data handling: JWST/NIRSpec (Slit Spectroscopy), VLT/MUSE (IFU), Keck/NIRES (Echellette Spectrometer), SDSS-MaNGA (IFU)

OTHER PUBLICATIONS

[4] Rashmi Gottumukkala, Gabriel Brammer, Sune Toft, et al. "The DAWN JWST Archive: A spectroscopically-informed population study of the physical properties of galaxies at $z > 2$". In: *arXiv e-prints* (Oct. 11, 2026). arXiv: 2610.15901.

[5] Yuanze Luo, Justin S. Spilker, David J. Setton, et al. "SQuIGG$\vec{L}$E: Ubiquitous Cool Gas Outflows with Large Mass-Loading Factors in $z \sim 0.7$ Post-starburst Galaxies". In: *arXiv e-prints* (Sept. 25, 2026). arXiv: 2609.31308.

[6] David J. Setton, Jenny E. Greene, Rachel Bezanson, et al. "SQuIGG$\vec{L}$E: Spatially resolved NIRSpec Paα reveals that the most CO-luminous, $z \sim 0.7$ post-starburst galaxies exhibit falling—but still active—star formation behind optically thick dust". In: *arXiv e-prints* (Sept. 25, 2026). arXiv: 2609.31307.

[7] Gianluca Scarpe, Francesco Valentino, Kei Ito, et al. "Morphology, Sizes, and Scatter in a Large Sample of Distant Quiescent Galaxies". In: *Astronomy & Astrophysics* 711 (July 2026), A177. DOI: 10.1051/0004-6361/202659598.

[8] K. Ito, F. Valentino, G. Brammer, et al. "DeepDive: A Deep Dive into the Physics of the First Massive Quiescent Galaxies in the Universe". In: *Astronomy & Astrophysics* 710 (June 2026), A364. DOI: 10.1051/0004-6361/202556137.

[9] F. Valentino, A. Pensabene, A. Weibel, et al. "Extended [CII] gas emission in and around a massive quiescent galaxy at $z = 7.3$". In: *arXiv e-prints* (June 19, 2026). arXiv: 2606.21361.

[10] F. Valentino, K. Ito, M. Farcy, et al. "High-ionization coronal lines trace quasar-like activity in recently quenched galaxies at high redshift". In: *arXiv e-prints* (May 28, 2026). arXiv: 2605.30424.

[11] Takumi Kakimoto, Masayuki Tanaka, Kei Ito, et al. "DeepDive: Simultaneous Formation of Massive Quiescent Galaxies in High-Redshift Galaxy Overdensities". In: *arXiv e-prints* (Apr. 22, 2026). arXiv: 2604.21007.

[12] William M. Baker, Kei Ito, Francesco Valentino, et al. "Double Trouble: Two Spectroscopically Confirmed Low-Mass Quiescent Galaxies at z $>$ 5 in Overdensities". In: *Astronomy & Astrophysics* 706 (Feb. 2026), A91. DOI: 10.1051/0004-6361/202557207.

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[13] Massissilia L. Hamadouche, Katherine E. Whitaker, Francesco Valentino, et al. “DeepDive: Tracing the early quenching pathways of massive quiescent galaxies at $z > 3$ from their star-formation histories and chemical abundances”. In: *arXiv e-prints* (Feb. 2, 2026). arXiv: 2602.02485.

[14] David J. Setton, Justin S. Spilker, Rachel Bezanson, et al. “SQuIGG$\vec{L}$E: Buried Star Formation Cannot Explain the Rapidly Fading CO(2-1) Luminosity in Massive, z $\sim$ 0.7 Post-starburst Galaxies”. In: *The Astronomical Journal* 170 (Dec. 2025), p. 351. DOI: 10.3847/1538-3881/ae1607.

[15] Vincenzo R. D’Onofrio, Justin S. Spilker, Rachel Bezanson, et al. “Quenching through Tidal Gas Removal: Molecular Gas and Star Formation in Tidal Tails of z $\sim$ 0.7 Post-starburst Galaxies”. In: *The Astrophysical Journal* 990.2 (Sept. 2025), p. 166. DOI: 10.3847/1538-4357/adf5bd.

[16] F. Valentino, K. E. Heintz, G. Brammer, et al. “Gas Outflows in Two Recently Quenched Galaxies at z = 4 and 7”. In: *Astronomy & Astrophysics* 699 (July 2025), A358. DOI: 10.1051/0004-6361/202553908.

[17] K. Ito, F. Valentino, M. Farcy, et al. “A Merging Pair of Massive Quiescent Galaxies at z = 3.44 in the Cosmic Vine”. In: *Astronomy & Astrophysics* 697 (May 2025), A111. DOI: 10.1051/0004-6361/202453211.

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